

## Lecture 2

# Differential forms

**Objectives.** *A calculus lecture, and the only one in this course. We need one number,  $\oint_{\gamma} p \cdot dq$ , to depend on the loop  $\gamma$  only through which way it winds — not on how fast it is traversed, not on the coordinates it is written in, and not on the particular loop within its class. Building the machinery that makes that true takes one lecture, and it pays for the rest of Part I: the symplectic form of [Lecture 3](#), the period lattice of [Lecture 4](#), and the action variables of [Lecture 5](#) are all written in the language assembled here.*

## 2.1 Why forms: 1-forms and the cotangent space

### The object we are trying to build

In [Lecture 5](#) we shall meet the integral

$$I = \frac{1}{2\pi} \oint_{\gamma} p \cdot dq = \frac{1}{2\pi} \oint_{\gamma} \sum_{i=1}^n p_i dq_i, \quad (2.1)$$

taken around a closed loop  $\gamma$  on an invariant torus, and everything in [Lectures 4](#) and [5](#) depends on it being a well-defined number. Three separate things have to be true, and not one of them is obvious:

1. The integrand must be an object that a *curve* can integrate, with no extra structure supplied. In particular no length, no angle, no inner product: phase space has no metric, and any construction that secretly uses one is not available to us.
2. The answer must not change when the curve is reparametrised. A loop traversed quickly and the same loop traversed slowly must give the same number.
3. The answer must not change when  $\gamma$  is deformed. Otherwise (2.1) labels a curve rather than a torus, and there is no “action variable” at all.

The first two are settled in §§ 2.1 and 2.5 by getting the *type* of the integrand right. The third is a theorem, and it is the real content of this lecture: it holds because  $p \cdot dq$  is *closed*, and the loops it is integrated over are not boundaries. Both words are defined below.

### Covectors

Throughout this lecture  $\mathcal{M}$  is a smooth manifold of dimension  $m$ , and  $x^1, \dots, x^m$  is a chart on it. In the application  $\mathcal{M}$  is phase space and  $m = 2n$ , with  $n$  the number of degrees of freedom —  $n$  is reserved for that throughout the course, so it is not available here.

Recall from § 1.2 that a tangent vector at  $p \in \mathcal{M}$  is the velocity of a curve through  $p$ , equivalently a derivation on functions near  $p$ , and that these form an  $m$ -dimensional vector space  $T_p\mathcal{M}$ . The chart gives  $T_p\mathcal{M}$  the basis  $\partial/\partial x^1, \dots, \partial/\partial x^m$ .

**Definition 2.1 (Covector, cotangent space).** A covector at  $p$  is a linear map  $\alpha_p: T_p\mathcal{M} \rightarrow \mathbb{R}$ . The set of them is the *cotangent space*  $T_p^*\mathcal{M}$ , the dual of  $T_p\mathcal{M}$ ; it is again a vector space of dimension  $m$ . A *differential 1-form*  $\alpha$  on  $\mathcal{M}$  assigns to each point  $p$  a covector  $\alpha_p$ , smoothly in  $p$ .

The example to hold on to is the differential of a function.

**Definition 2.2 (The differential of a function).** For  $f$  smooth near  $p$  and  $v \in T_p\mathcal{M}$ , set

$$df_p(v) = v(f), \quad (2.2)$$

the derivative of  $f$  along  $v$ . This is linear in  $v$ , so  $df_p \in T_p^*\mathcal{M}$ , and  $df$  is a 1-form.

Applying (2.2) to the coordinate functions  $x^i$  themselves gives  $dx^i(\partial/\partial x^j) = \partial x^i/\partial x^j = \delta_j^i$ : the 1-forms  $dx^1, \dots, dx^m$  are exactly the basis of  $T_p^*\mathcal{M}$  dual to the coordinate basis of  $T_p\mathcal{M}$ . Hence every 1-form is

$$\alpha = \sum_i a_i(x) dx^i, \quad \text{and in particular} \quad df = \sum_i \frac{\partial f}{\partial x^i} dx^i, \quad (2.3)$$

the second identity being (2.2) evaluated on each basis vector. So the notation  $df$ , which until now was a formal symbol in the chain rule, is a genuine object: a machine that eats a tangent vector and returns a number.

**Remark 2.3 (Why not a vector field?).** The components  $\partial f/\partial x^i$  are also the components of the gradient, so it is fair to ask what is gained. The answer is that the gradient is not defined by them. To turn the collection  $\partial f/\partial x^i$  into a *vector* one must raise an index, and that needs an inner product; under a change of coordinates the honest transformation law of  $\partial f/\partial x^i$  is the one covectors have, not the one vectors have. Phase space carries no inner product, so on  $\mathcal{M}$  the gradient of a function does not exist while its differential does. This is not fastidiousness: it is why  $p \cdot dq$  makes sense on  $\mathcal{M}$  at all.

### The picture: a stack of surfaces

A tangent vector is drawn as an arrow. A covector is not an arrow; it is a *stack*. Draw the level sets  $f = c, f = c + 1, f = c + 2, \dots$  near  $p$ : for a nearly-constant  $f$  they are far apart, for a rapidly varying  $f$  they are close together (Figure 2.1). Then (2.2) says

$df_p(v)$  is the number of level surfaces the arrow  $v$  pierces.

Figure 2.2 makes Remark 2.3 visible. Nothing in the counting used how long  $v$  is or what angle it makes with anything; a covector is exactly the structure that can be evaluated on a vector with no metric present. Addition of covectors is superposition of stacks (Figure 2.3):  $\alpha + \beta$  is denser than either.

**Example 2.4 (The canonical 1-form, met early).** On  $\mathcal{M} = T^*\mathbb{R}^n$  with coordinates  $q_1, \dots, q_n, p_1, \dots, p_n$ , the expression  $\sum_i p_i dq_i$  is a 1-form: at each point of phase space it eats a tangent vector  $v$  and returns  $\sum_i p_i dq_i(v)$ , the momenta contracted with the  $q$ -components of  $v$ . It is emphatically *not* of the form  $df$  — that is the subject of § 2.4, and the reason (2.1) is ever nonzero. Lecture 3 will show that this 1-form exists on any cotangent bundle without choosing coordinates.

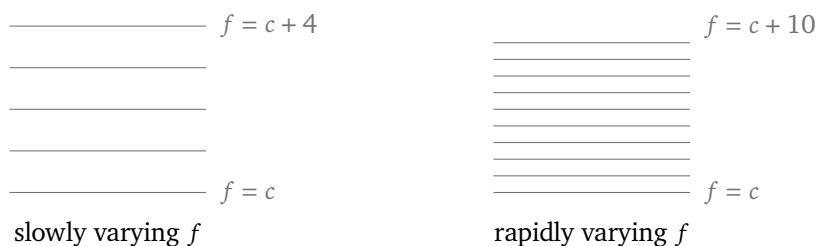


Figure 2.1: A covector is a stack, not an arrow. The same function drawn where it varies slowly and where it varies fast:  $df$  is *denser* where  $f$  changes quickly. Nothing in the picture requires a notion of length — only the surfaces and their labels.

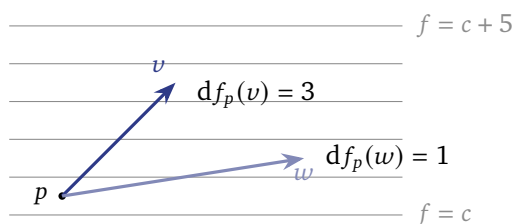


Figure 2.2:  $df_p(v)$  counts the level surfaces pierced by  $v$ . The long arrow  $w$  gives the smaller answer: it is the crossing of surfaces that is measured, not the length of the arrow.

## 2.2 2-forms, oriented area and the wedge product

### What we want to measure

A 1-form measures a curve's worth of information: one vector in, one number out. To integrate over a *surface* we need a machine taking two vectors, and it should return the area of the parallelogram they span — because that is what a surface integral adds up.

Start in the plane, where the answer is already known from linear algebra: the parallelogram spanned by  $v = (v^1, v^2)$  and  $w = (w^1, w^2)$  has signed area

$$A(v, w) = \det \begin{pmatrix} v^1 & w^1 \\ v^2 & w^2 \end{pmatrix} = v^1 w^2 - v^2 w^1. \quad (2.4)$$

Three properties of  $A$  are worth naming before we generalise, because they are forced by what “area” means and not chosen for convenience.

- *Bilinearity*. Doubling one edge doubles the area; gluing two parallelograms along an edge adds their areas.
- $A(v, v) = 0$ . A degenerate parallelogram has no area.
- $A(w, v) = -A(v, w)$ . The area is *signed*: it records not only how much area there is but which way round the two edges run.

The third property is the one students find arbitrary, so note that it is not an extra assumption: it follows from the first two. Expanding  $0 = A(v + w, v + w) = A(v, v) + A(v, w) + A(w, v) + A(w, w)$  gives  $A(v, w) = -A(w, v)$  at once. *Antisymmetry is what you get for free once you ask that a degenerate parallelogram have zero area.*

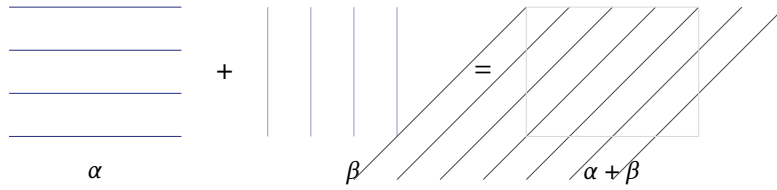


Figure 2.3: Adding covectors superposes stacks. The sum runs in a new direction and is denser than either summand, which is the geometric content of  $(\alpha + \beta)(v) = \alpha(v) + \beta(v)$ .



Figure 2.4: The same parallelogram, the same area, opposite orientations. A 2-form remembers the circulation arrow, and that memory is exactly the minus sign in  $\alpha \wedge \beta = -\beta \wedge \alpha$ .

**Definition 2.5 (2-form).** A 2-form at  $p$  is a bilinear map  $\beta_p: T_p\mathcal{M} \times T_p\mathcal{M} \rightarrow \mathbb{R}$  with  $\beta_p(v, w) = -\beta_p(w, v)$ . A differential 2-form assigns such a  $\beta_p$  to each point, smoothly.

### The wedge product

Two 1-forms produce a 2-form in exactly one natural way.

**Definition 2.6 (Wedge product of 1-forms).** For 1-forms  $\alpha, \beta$ ,

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v) = \det \begin{pmatrix} \alpha(v) & \alpha(w) \\ \beta(v) & \beta(w) \end{pmatrix}. \quad (2.5)$$

Antisymmetry in  $(v, w)$  is manifest, so  $\alpha \wedge \beta$  is a 2-form; and swapping the two *forms* transposes the determinant, so

$$\alpha \wedge \beta = -\beta \wedge \alpha, \quad \text{and in particular} \quad \alpha \wedge \alpha = 0. \quad (2.6)$$

This is the same minus sign as Figure 2.4, now in the algebra. It is worth saying out loud that (2.6) is a statement about *orientation*, not a convention about symbols: a plane with a circulation arrow drawn on it is a different object from the same plane with the arrow reversed, and  $\wedge$  is the product that knows the difference.

**Example 2.7 ( $dx \wedge dy$  is the signed area).** Take  $\alpha = dx$ ,  $\beta = dy$  in the plane. Then  $(dx \wedge dy)(v, w) = v^1 w^2 - v^2 w^1$ , which is precisely (2.4): see Figure 2.5. The abstract definition has reproduced the determinant, and we may now stop writing  $A$ .

In  $m$  dimensions, a 2-form has components  $\beta_{ij} = \beta(\partial_i, \partial_j)$  with  $\beta_{ij} = -\beta_{ji}$ , so

$$\beta = \sum_{i < j} \beta_{ij} dx^i \wedge dx^j, \quad (2.7)$$

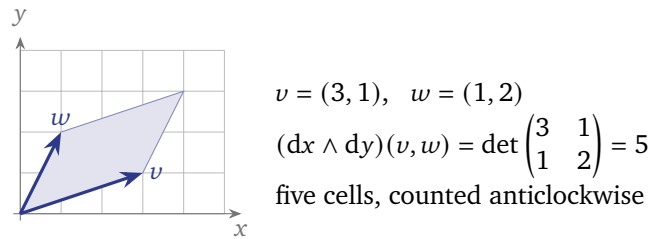


Figure 2.5:  $dx \wedge dy$  evaluated on two vectors. The number returned is the signed area of the parallelogram they span, in units of the cell  $\partial_x, \partial_y$  — positive here because  $v, w$  run anticlockwise.

a sum over  $\binom{m}{2}$  independent components. In the plane there is one,  $dx \wedge dy$ ; in  $\mathbb{R}^3$  there are three; in  $\mathbb{R}^4$ , six — and it is the  $\mathbb{R}^4$  count that [Lecture 3](#) will need when it writes  $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ .

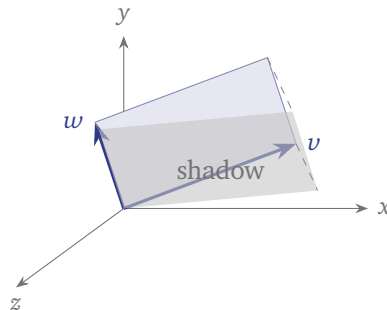


Figure 2.6:  $(dx \wedge dy)(v, w)$  is the signed area not of the parallelogram but of its shadow in the  $xy$ -plane. A 2-form does not measure area; it measures area *as seen in a chosen pair of directions*, which is why there is one 2-form per pair of coordinates.

### Higher degree, and the one rule we need

The pattern of (2.5) continues. A  $k$ -form is a multilinear map on  $k$  tangent vectors that changes sign whenever two arguments are swapped, and the wedge of  $k$  1-forms is the determinant of the matrix of pairings:

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) = \det[\alpha^i(v_j)]_{i,j=1}^k. \quad (2.8)$$

We need this course only in the two places where it is unavoidable.

- *Top degree in  $\mathbb{R}^3$ .* There is exactly one 3-form up to scale,  $dx \wedge dy \wedge dz$ , and (2.8) says it returns the signed volume of the parallelepiped on  $u, v, w$ . Without it the statement “d of a 2-form in  $\mathbb{R}^3$ ” has nowhere to live, and the divergence would not appear in § 2.3.
- *Top degree in  $\mathbb{R}^{2n}$ .* [Lecture 3](#) forms  $\omega^n = \omega \wedge \cdots \wedge \omega$  and finds the Liouville volume of § 1.6 sitting inside it.

One thing does have to be said, because the second bullet uses it and (2.8) does not supply it: *what  $\omega \wedge \omega$  means*, when  $\omega$  is a 2-form and not a 1-form.

**Definition 2.8 (Wedge in general).** The wedge product is bilinear and associative, and on products of 1-forms it simply concatenates the factors,

$$(\alpha^1 \wedge \cdots \wedge \alpha^k) \wedge (\beta^1 \wedge \cdots \wedge \beta^\ell) = \alpha^1 \wedge \cdots \wedge \alpha^k \wedge \beta^1 \wedge \cdots \wedge \beta^\ell.$$

Since every form is a sum of such products of coordinate differentials, those two rules determine it. Moving one factor past another costs a sign each time, so in degrees  $k$  and  $\ell$

$$\eta \wedge \xi = (-1)^{k\ell} \xi \wedge \eta. \quad (2.9)$$

In particular a form of *odd* degree wedged with itself vanishes, while one of *even* degree need not — which is the only reason  $\omega^n$  is not zero.

Reading

**Example 2.9 ( $\omega^2$ , computed).** With  $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$  on  $\mathbb{R}^4$ , expanding bilinearly gives four terms. The two like terms vanish, because  $dq_i \wedge dp_i \wedge dq_i \wedge dp_i$  repeats a factor, and the two cross terms are equal by (2.9) with  $k = \ell = 2$ . So

$$\omega \wedge \omega = 2 dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2, \quad \text{that is} \quad \frac{\omega^2}{2!} = dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2.$$

The factorial in [Lecture 3](#)'s  $\omega^n/n!$  is there to cancel exactly this count, and nothing more.

Beyond this the general theory of  $k$ -forms is not developed here; (2.8), [Definition 2.8](#) and (2.9) are the whole of it that this course uses, and they are enough.

**Remark 2.10 (Degrees run out).** On an  $m$ -dimensional manifold every  $k$ -form with  $k > m$  vanishes: more than  $m$  vectors in an  $m$ -dimensional space are linearly dependent, and an alternating multilinear map vanishes whenever its arguments are — for a wedge of 1-forms this is (2.8) with a dependent set of columns. So in the plane 2-forms are top degree, and any 3-form is zero — a fact used without comment every time a computation in  $\mathbb{R}^2$  stops after one step.

## 2.3 The exterior derivative

We have  $d$  on functions. Extending it to forms of every degree is what turns this algebra into a calculus.

**Definition 2.11 (Exterior derivative).** On a 0-form (a function),  $df$  is [Definition 2.2](#). On a 1-form  $\alpha = \sum_i a_i dx^i$ ,

$$d\alpha = \sum_i da_i \wedge dx^i = \sum_{i,j} \frac{\partial a_i}{\partial x^j} dx^j \wedge dx^i, \quad (2.10)$$

and in general  $d$  acts on  $\sum a_{i_1 \dots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$  by differentiating each coefficient and wedging  $d$  of it on the left. It raises degree by one and obeys the graded Leibniz rule

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta, \quad \alpha \text{ of degree } k. \quad (2.11)$$

The formula is written in coordinates, so it is fair to ask whether a different chart would give a different answer. It would not:  $d$  is characterised by (2.11), by agreeing with

**Definition 2.2** in degree zero, and by  $d \circ d = 0$ , and those three conditions mention no coordinates. We take the characterisation on trust (*cited*; da Silva, §6) and use the formula.

**Example 2.12** ( $d$  in the plane is the circulation). For  $\alpha = P dx + Q dy$  on  $\mathbb{R}^2$ , (2.10) gives

$$d\alpha = \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy, \quad (2.12)$$

using  $dx \wedge dx = 0$  and  $dy \wedge dx = -dx \wedge dy$ . The coefficient is the curl, and Figure 2.7 says what it measures:  $d\alpha$  evaluated on a small parallelogram is the circulation of  $\alpha$  around its boundary, per unit area. That reading is the whole of Stokes' theorem in miniature, and § 2.5 does no more than add such squares up.

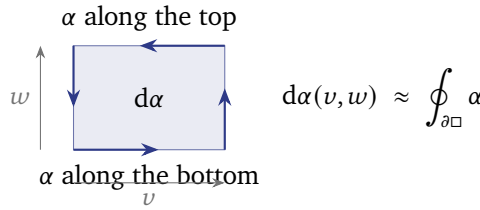


Figure 2.7:  $d\alpha$  on an infinitesimal parallelogram is the circulation of  $\alpha$  around its boundary. The four edge contributions do not cancel exactly, and what is left over is  $d\alpha$ .

**Theorem 2.13** ( $d \circ d = 0$ ).  $d(d\alpha) = 0$  for every form  $\alpha$ .

*Proof.* It is enough to check it on a function, since the general case follows by (2.11) and  $d(dx^i) = 0$ . With  $\alpha = f$ ,

$$d(df) = d\left(\sum_i \frac{\partial f}{\partial x^i} dx^i\right) = \sum_{i,j} \frac{\partial^2 f}{\partial x^j \partial x^i} dx^j \wedge dx^i.$$

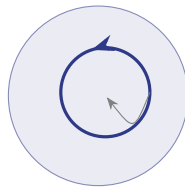
Now split the double sum into the pairs  $\{i, j\}$ . Within each pair the coefficient  $\partial^2 f / \partial x^j \partial x^i$  is symmetric in  $i, j$  because mixed partials of a smooth function commute, while  $dx^j \wedge dx^i$  is antisymmetric by (2.6). Within each pair the two terms are equal and opposite, so the sum vanishes pair by pair. ■

**Definition 2.14** (Closed, exact). A  $k$ -form  $\alpha$  is closed if  $d\alpha = 0$ , and exact if  $\alpha = d\beta$  for some  $(k-1)$ -form  $\beta$ .

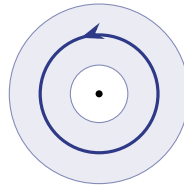
Theorem 2.13 says exact  $\Rightarrow$  closed. The converse is the interesting direction, and it is local.

**Theorem 2.15** (Poincaré lemma · cited). A closed form on a ball — or on any region that can be smoothly contracted to a point — is exact. (da Silva, §6; Arnold, §36.)

Everything this course does with forms lives in the gap between Theorem 2.13 and Theorem 2.15: closed and exact differ, and they differ only because of the shape of the region (Figure 2.8). § 2.4 is the smallest example, and Lecture 5 is the payoff.



a disc  
closed = exact



an annulus  
closed  $\not\supseteq$  exact

The forms do not change; the region does. On the left every loop shrinks to a point and [Theorem 2.15](#) applies. On the right one loop cannot, and (2.13) lives in the difference.

Figure 2.8: Closed versus exact is a statement about the *region*. Locally the two notions agree; globally they are separated by exactly the loops that fail to bound.

### Reading

## Grad, curl and div are one operator

In  $\mathbb{R}^3$  the forms of each degree have three, three and one independent components — and so do the classical vector operators. Writing  $d$  in each degree reproduces them:

| degree | object                                                  | $d$ gives                                                                | classical name |
|--------|---------------------------------------------------------|--------------------------------------------------------------------------|----------------|
| 0      | $f$                                                     | $\sum_i \partial_i f \, dx^i$                                            | grad           |
| 1      | $Pdx + Qdy + Rdz$                                       | a 2-form with components $\partial_y R - \partial_z Q, \dots$            | curl           |
| 2      | $Pdy \wedge dz + Q \, dz \wedge dx + R \, dx \wedge dy$ | $(\partial_x P + \partial_y Q + \partial_z R) \, dx \wedge dy \wedge dz$ | div            |

Read in this table, [Theorem 2.13](#) is two familiar identities at once:  $\text{curl grad} = 0$  and  $\text{div curl} = 0$ . They were never two facts. The dictionary is special to three dimensions — it relies on 3-forms and 1-forms both having three components — which is the reason the classical operators do not generalise and  $d$  does. Phase space is  $2n$ -dimensional, so from here on only  $d$  is available — which is why this comparison is worth making once and then leaving behind.

## 2.4 Closed but not exact: the annulus

**Example 2.16** (The angle form). On the punctured plane  $\mathcal{M} = \mathbb{R}^2 \setminus \{0\}$  — an annulus, for present purposes — put

$$\alpha = \frac{x \, dy - y \, dx}{x^2 + y^2}. \quad (2.13)$$

Away from the origin this is smooth, and a two-line computation with (2.12) gives  $d\alpha = 0$ : it is *closed*.

In polar coordinates  $\alpha = d\theta$ , which makes it look exact. It is not, and the reason is that  $\theta$  is not a function on the annulus: continuing it once around the hole returns it increased by  $2\pi$ . The name  $d\theta$  is a name for the 1-form (2.13), not an assertion that something has been differentiated. To see this is not a quibble, integrate.

**Proposition 2.17.**  $\alpha$  is closed and not exact on  $\mathbb{R}^2 \setminus \{0\}$ .

*Proof.* Closedness is [Example 2.16](#). For the rest, let  $\gamma$  be the unit circle traversed once anticlockwise,  $\gamma(t) = (\cos t, \sin t)$ ,  $t \in [0, 2\pi]$ . Substituting into (2.13) gives  $\oint_{\gamma} \alpha = \int_0^{2\pi} dt = 2\pi$ . If  $\alpha$  were  $df$  for a function  $f$  on the annulus, then by [Proposition 2.22](#) below the integral around any closed curve would be  $f(\gamma(2\pi)) - f(\gamma(0)) = 0$ . It is  $2\pi$ , so no such  $f$  exists. ■

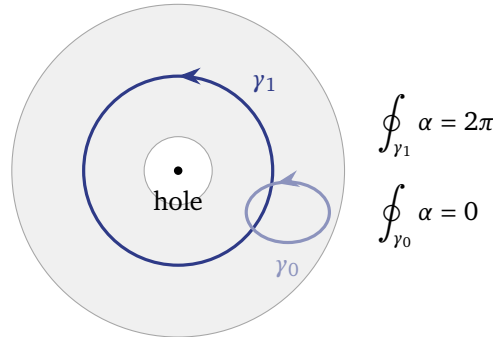


Figure 2.9: Two loops in the annulus.  $\gamma_0$  can be shrunk to a point inside the region and integrates to zero;  $\gamma_1$  cannot, because the hole is in the way, and integrates to  $2\pi$ . The obstruction to exactness is the hole, and nothing else.

This is the smallest interesting fact in the lecture, and the one to remember. It is tempting to read [Proposition 2.17](#) as bookkeeping. It is not:

*$p \cdot dq$  is closed on an invariant torus, which is what makes (2.1) depend on the loop only through its class; and it need not be exact there, which is what allows the answer to be nonzero.*

The two halves do different jobs and neither can be dropped. Were  $p \cdot dq$  exact on the torus, (2.1) would give  $I = 0$  for every cycle on it, by [Proposition 2.22](#). The nonzero actions of the oscillator, the pendulum and the Kepler problem exist for the reason [Figure 2.9](#) exhibits — a cycle that does not bound, and a closed form with no primitive along it.

Reading

**Remark 2.18 (Not a universal claim).** “Not exact” is what makes a nonzero action possible, not something that holds on every invariant torus. On  $T^*S^1$  with  $H = p$ , the level  $p = 0$  is a perfectly good compact invariant circle on which  $p \, dq$  restricts to 0 — exact, with action zero, and no contradiction anywhere. What [Lecture 5](#) needs from this lecture is the *closedness*, which is what makes the period well defined at all.

## 2.5 Pullback, integration, and Stokes’ theorem

### Pullback

Forms have one property vectors do not: they can be transported *backwards* along any smooth map, with no assumption that the map be invertible.

**Definition 2.19 (Pullback).** Let  $F: N \rightarrow M$  be smooth and let  $\alpha$  be a 1-form on  $M$ . Its pullback  $F^*\alpha$  is the 1-form on  $N$  given by

$$(F^*\alpha)_p(v) = \alpha_{F(p)}(dF_p v), \quad (2.14)$$

where  $dF_p: T_p N \rightarrow T_{F(p)} M$  is the derivative of  $F$ . For a 2-form,  $(F^*\beta)_p(v, w) = \beta_{F(p)}(dF_p v, dF_p w)$ , and similarly in any degree.

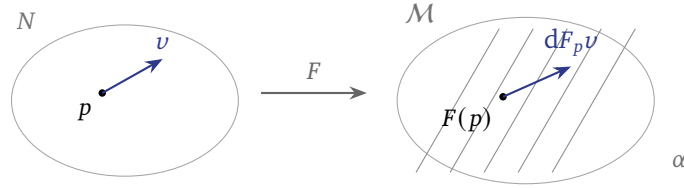


Figure 2.10: Pullback. The vector is pushed forward by  $dF$ , the form is evaluated where it lives, and the answer is recorded back at  $p$ . Nothing needs to be inverted, which is why  $F^*$  exists for every smooth  $F$ .

In practice pullback is substitution: if  $F$  is given by  $x^i = x^i(u)$ , then  $F^*(dx^i) = \sum_a (\partial x^i / \partial u^a) du^a$ , and one simply substitutes. Two consequences are worth recording, both because they are used later and because they are the reason the formalism is worth its cost:

$$F^*(df) = d(f \circ F), \quad F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta, \quad F^*(d\alpha) = d(F^*\alpha). \quad (2.15)$$

The last of these —  $d$  commutes with pullback — is what makes “closed” and “exact” coordinate-independent notions, and [Lecture 3](#) will use the middle one to recognise canonical transformations as the maps satisfying  $F^*\omega = \omega$ .

### Integration over a curve

Now the type-checking of [§ 2.1](#) pays off.

**Definition 2.20** (Line integral of a 1-form). For a smooth curve  $\gamma: [a, b] \rightarrow M$  and a 1-form  $\alpha$ ,

$$\int_\gamma \alpha = \int_a^b \gamma^* \alpha = \int_a^b \alpha_{\gamma(t)}(\dot{\gamma}(t)) dt. \quad (2.16)$$

**Proposition 2.21** (Reparametrisation invariance). If  $\varphi: [c, d] \rightarrow [a, b]$  is smooth, increasing and onto, then  $\int_{\gamma \circ \varphi} \alpha = \int_\gamma \alpha$ .

*Proof* (★). By the chain rule  $\frac{d}{ds}(\gamma \circ \varphi) = \dot{\gamma}(\varphi(s)) \varphi'(s)$ , so the integrand acquires exactly the factor  $\varphi'(s)$  that the substitution rule removes. ■

So the answer depends on the curve and its direction, not on the schedule. That was requirement 2 of [§ 2.1](#), and [Definition 2.20](#) met it by construction: a 1-form is precisely the kind of object whose integral over a curve needs no further data.

**Proposition 2.22** (Fundamental theorem).  $\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$ . In particular  $\oint_\gamma df = 0$  for every closed curve.

*Proof*.  $df(\dot{\gamma}) = \frac{d}{dt}(f \circ \gamma)$  by (2.2), and the one-variable fundamental theorem of calculus does the rest. ■

[Proposition 2.22](#) is what [Proposition 2.17](#) used, and it is also the statement that an exact form carries no information about loops. Everything interesting therefore lies with forms that are closed and not exact.

## Stokes' theorem

**Theorem 2.23** (Stokes · cited). Let  $S$  be a compact oriented surface with boundary  $\partial S$ , carrying the induced orientation, and let  $\alpha$  be a 1-form defined on a neighbourhood of  $S$ . Then

$$\int_S d\alpha = \oint_{\partial S} \alpha. \quad (2.17)$$

The same identity holds in every degree, for a  $k$ -form on a  $(k+1)$ -dimensional region. (Proof: da Silva, §6; Arnold, §36.)

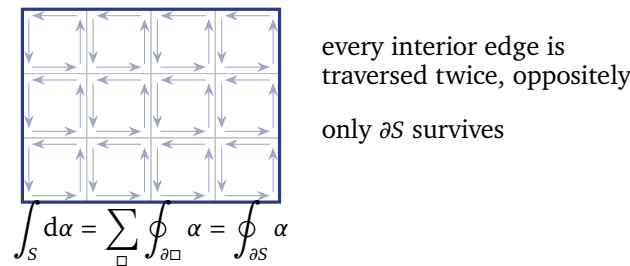
We do not prove it. What we do instead is say exactly why it is true, because the mechanism is short and is the picture the rest of Part I uses.

Reading

**Example 2.24** (Stokes on a rectangle, verified). Let  $S = [x_0, x_1] \times [y_0, y_1]$  and  $\alpha = P dx + Q dy$ . By (2.12) and Fubini,

$$\int_S d\alpha = \int_{y_0}^{y_1} \int_{x_0}^{x_1} \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_{y_0}^{y_1} [Q]_{x_0}^{x_1} dy - \int_{x_0}^{x_1} [P]_{y_0}^{y_1} dx,$$

after doing one integral in each term by the fundamental theorem of calculus. The four resulting boundary terms are exactly the four edges of  $\oint_{\partial S} \alpha$ , each with the sign the anticlockwise orientation gives it. That is (2.17) for a rectangle.



every interior edge is  
traversed twice, oppositely  
only  $\partial S$  survives

Figure 2.11: Why Stokes is true. Chop  $S$  into cells; on each,  $d\alpha$  is the circulation around its boundary (Figure 2.7). Summing, every interior edge is traversed once in each direction and cancels, leaving the outer boundary alone.

**Remark 2.25** (What we owe, and to whom · cited). Two results here are cited rather than proved — Theorem 2.15 and Theorem 2.23 — but only one of them is *used* by a later proof, and it is used in the strong sense: Lecture 5 proves that the action variables (2.1) are well defined *using Stokes*. That proof is therefore *proved modulo Stokes*, and Lecture 5 says so rather than claiming more than it has. This is the course's rigour convention at work: a result proved with a cited tool inherits the qualification.

## The payoff: integrals that depend only on the class of the loop

**Corollary 2.26** (Deformation invariance). Let  $\alpha$  be a closed 1-form on a region  $U$ , and let  $F: S^1 \times [0, 1] \rightarrow U$  be a smooth deformation — a family of closed curves  $\gamma_s = F(\cdot, s)$  inside

$U$ , running from  $\gamma_0$  to  $\gamma_1$ . Then

$$\oint_{\gamma_1} \alpha = \oint_{\gamma_0} \alpha.$$

*Proof.* Pull  $\alpha$  back to the cylinder  $C = S^1 \times [0, 1]$ , where it is a 1-form on a 2-dimensional region. It is still closed, because  $d$  commutes with pullback (2.15):  $d(F^*\alpha) = F^*(d\alpha) = 0$ . The boundary of  $C$  is its two ends, the top with the orientation induced from  $C$  and the bottom with the opposite one, so  $\partial C = S^1 \times \{1\} - S^1 \times \{0\}$ . By Theorem 2.23 applied on  $C$ ,

$$\oint_{\gamma_1} \alpha - \oint_{\gamma_0} \alpha = \oint_{\partial C} F^*\alpha = \int_C d(F^*\alpha) = 0. \quad \blacksquare$$

**Remark 2.27** (Why the statement is about a deformation, not a shape). It is tempting to say instead “if  $\gamma_0$  and  $\gamma_1$  together bound a surface, the integrals agree”. That is false as stated, and the annulus supplies the counterexample: the circles  $r = 1$  and  $r = 2$  together bound the region between them, but if  $\gamma_1$  is  $r = 2$  anticlockwise and  $\gamma_0$  is  $r = 1$  clockwise, then  $\oint_{\gamma_1} \alpha = 2\pi$  while  $\oint_{\gamma_0} \alpha = -2\pi$ . Bounding is a statement about point sets; what Stokes consumes is an *oriented* boundary, and a deformation supplies the orientations along with the surface. This is also why  $F$  may be any smooth map: a wildly self-crossing loop does not sweep out an embedded surface, and the proof above never needs one.

**Example 2.28** (The annulus again, quantitatively). Take  $\alpha$  as in (2.13). Any loop that winds once anticlockwise about the hole can be deformed to the unit circle *without leaving the annulus*, so by Corollary 2.26 every one of them gives  $2\pi$  — the circle of radius 1 and a wildly self-crossing loop alike. A loop winding  $m$  times gives  $2\pi m$ , and the small loop  $\gamma_0$  of Figure 2.9 shrinks to a constant curve, so it gives 0. The integral sees nothing about the loop except its winding number.

The hypothesis is doing real work, and it is worth seeing where. To deform  $\gamma_1$  to a point, the family of curves would at some stage have to cross the puncture, where  $\alpha$  is not defined — so there is no such  $F$  into  $U$ , Corollary 2.26 does not apply, and the two answers are free to differ. They do:  $2\pi$  against 0.

This is requirement 3 of § 2.1, discharged. An action variable is attached not to a curve but to a *class* of curves, and Lecture 4 will produce exactly  $n$  such classes on an  $n$ -dimensional invariant torus — the basis of cycles  $\gamma_1, \dots, \gamma_n$  that Figure 2.12 shows for  $n = 2$ , and Lecture 5 will integrate  $p \cdot dq$  over each of them in turn.

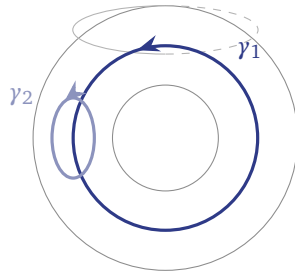


Figure 2.12: The two independent cycles on  $T^2$ , drawn on the torus seen from above. Neither bounds, so a closed 1-form may integrate to something nonzero over each, and Lecture 5 calls the two answers  $2\pi I_1$  and  $2\pi I_2$ . This picture is the target of the whole lecture; Lecture 4 supplies the torus.

### Summary.

- *1-forms*. A covector eats a tangent vector and returns a number, with no metric involved (Definition 2.1);  $df$  is the model (Definition 2.2), and the picture is a stack of level surfaces that a vector pierces (Figure 2.2).
- *2-forms and  $\wedge$* . Antisymmetry is forced by asking that a degenerate parallelogram have zero area, not imposed (§ 2.2);  $dx \wedge dy$  is the signed area (Example 2.7); a wedge of 1-forms is a determinant (2.8).
- *The exterior derivative*. One operator  $d$  (Definition 2.11) with  $d \circ d = 0$  (Theorem 2.13); in  $\mathbb{R}^3$  it is grad, curl and div, and  $d \circ d = 0$  is curl grad = 0 and div curl = 0 at once.
- *Closed versus exact*. Exact  $\Rightarrow$  closed always; the converse holds locally (Theorem 2.15) and fails globally. The angle form on the annulus is the whole story in one example (Proposition 2.17).
- *Stokes, and what it buys*.  $\int_S d\alpha = \oint_{\partial S} \alpha$  (Theorem 2.23, cited); hence the integral of a closed form around a loop is unchanged by any deformation that stays inside the region (Corollary 2.26). That corollary is what makes  $I = \frac{1}{2\pi} \oint_\gamma p \cdot dq$  a function of the torus rather than of the curve.

**Tutorial.** Exercise Sheet 2.

## References

da Silva, *Lectures on Symplectic Geometry*, §6 (forms,  $d$ , the Poincaré lemma, Stokes); Arnold, *Mathematical Methods of Classical Mechanics*, §§32, 35–36 (exterior algebra, integration of forms, Stokes — and §44 for the action integral this lecture was built to support).